

1. A projectile is launched with an initial speed of 200 m/s at an angle of 60° with respect to the horizontal. Compute the range R as measured up the incline.

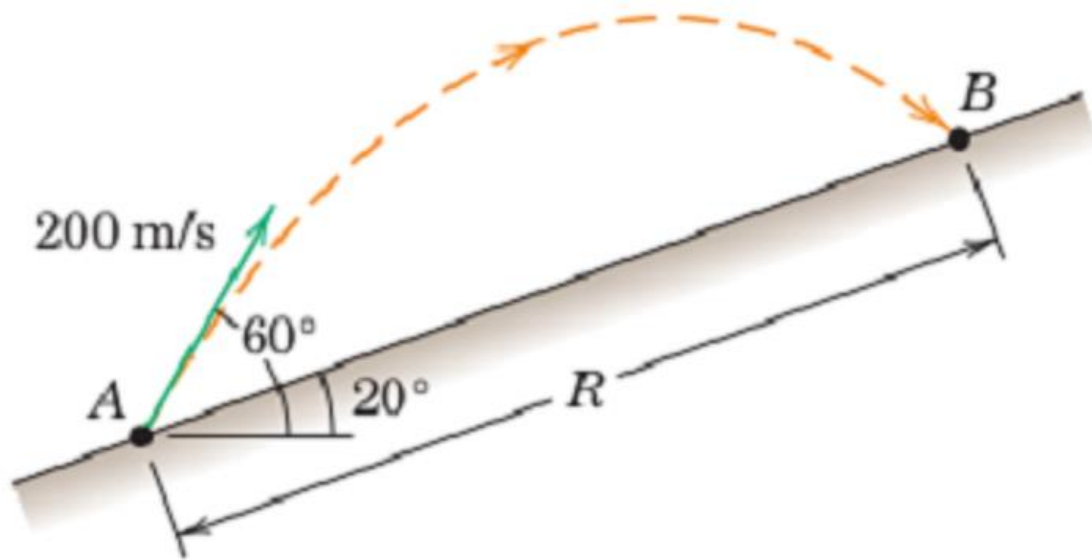


Figure: 1

2. To anticipate the dip and hump in the road, the driver of a car applies her brakes to produce a uniform deceleration. Her speed is 100 km/h at the bottom A of the dip and 50 km/h at the top C of the hump, which is 120 m along the road from A. If the passengers experiences a total acceleration of 3 m/sec^2 at A and if the radius of curvature of the hump at C is 150 m, calculate (a) the radius of curvature ρ at A, (b) the acceleration at the inflection point B, and (c) the total acceleration at C.

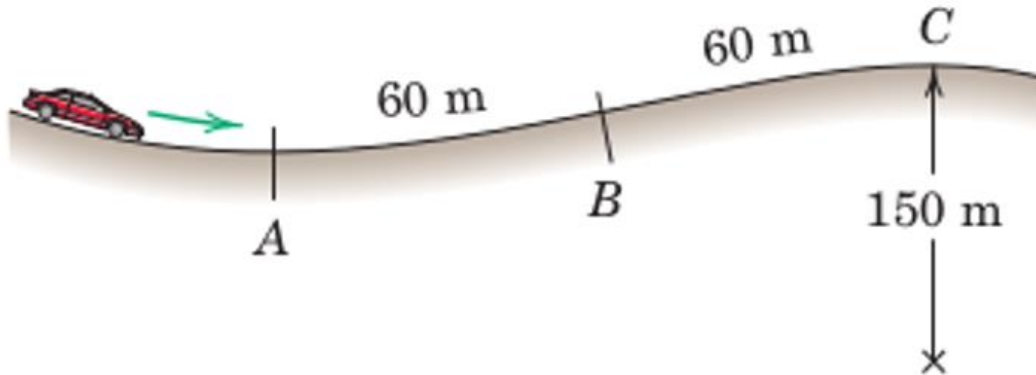


Figure: 2

3. Write the vector expression for the acceleration $\mathbf{a}$ of the mass center G of the simple pendulum in both n - t and x - y coordinates for the instant when $\theta = 60^\circ$ if $\dot{\theta} = 2 \text{ rad/sec}$ and $\ddot{\theta} = 4.025 \text{ rad/sec}^2$.

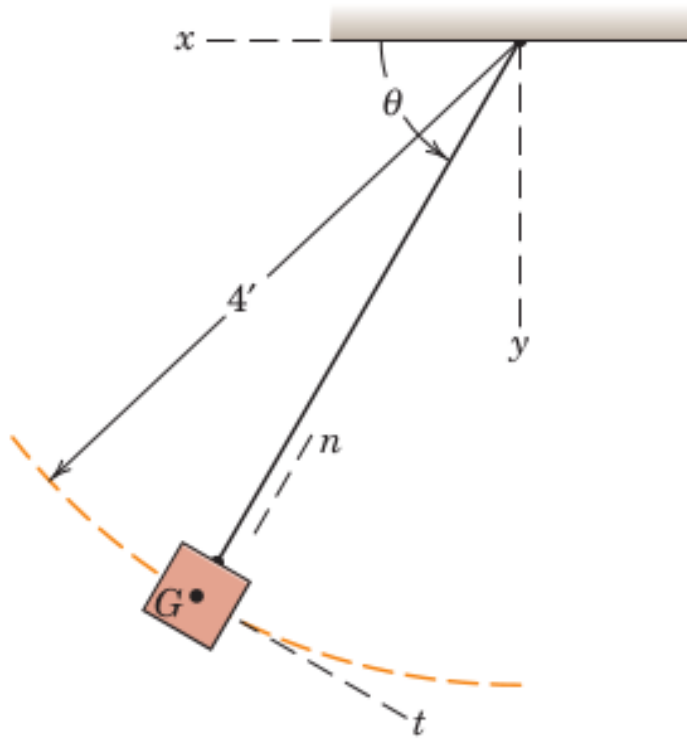


Figure: 3

4. The cam is designed so that the centre of the roller A which follows the contour moves on the limaçon defined by $r = b - c \cos \theta$, where $b > c$. If the cam does not rotate, determine the magnitude of the total acceleration of A in terms of θ if the slotted arm revolves with a constant counter clockwise angular rate $\dot{\theta} = \omega$.

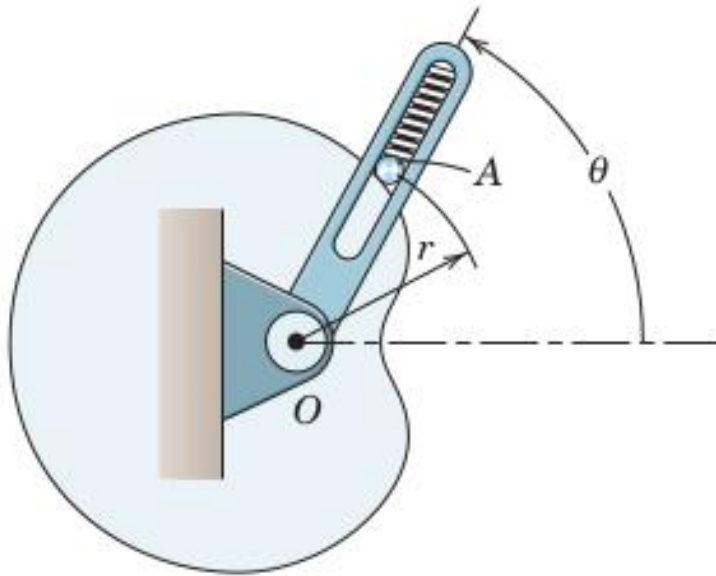


Figure: 4

5. A jet plane flying at a constant speed v at an altitude $h = 10$ km is being tracked by radar located at O directly below the line of flight. If the angle θ is decreasing at the rate of 0.020 rad/s when $\theta = 60^\circ$, determine the value of $\dot{r}$ at this instant and the magnitude of the velocity $\mathbf{v}$ of the plane.

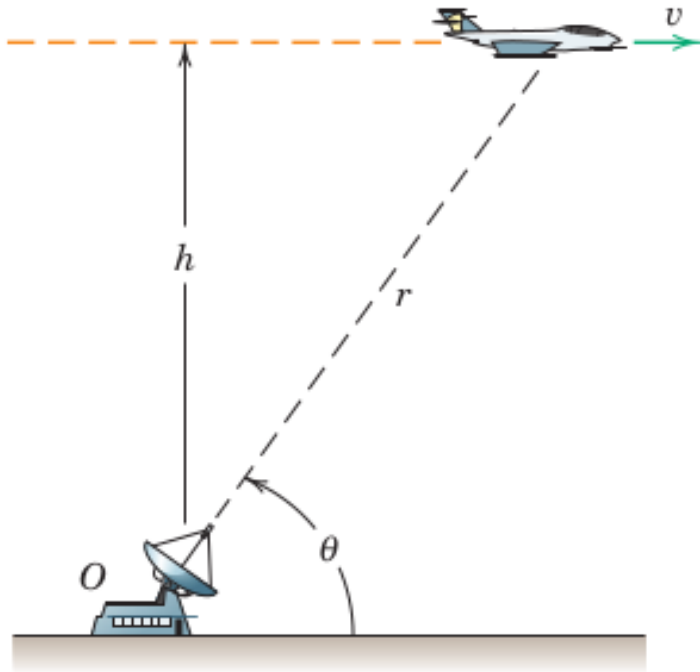


Figure:5

7. For the instant represented, car A is rounding the circular curve at the constant speed of 30 mi/hr, while car B is slowing down at the rate of 5 mi/hr per second. Determine the magnitude of the acceleration that car A appears to have to an observer in car B.

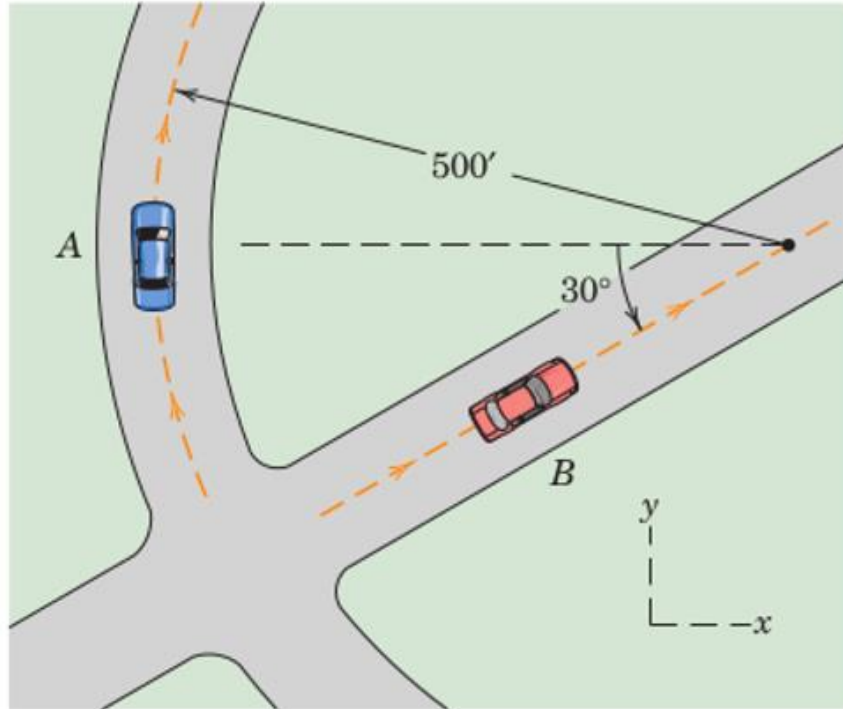
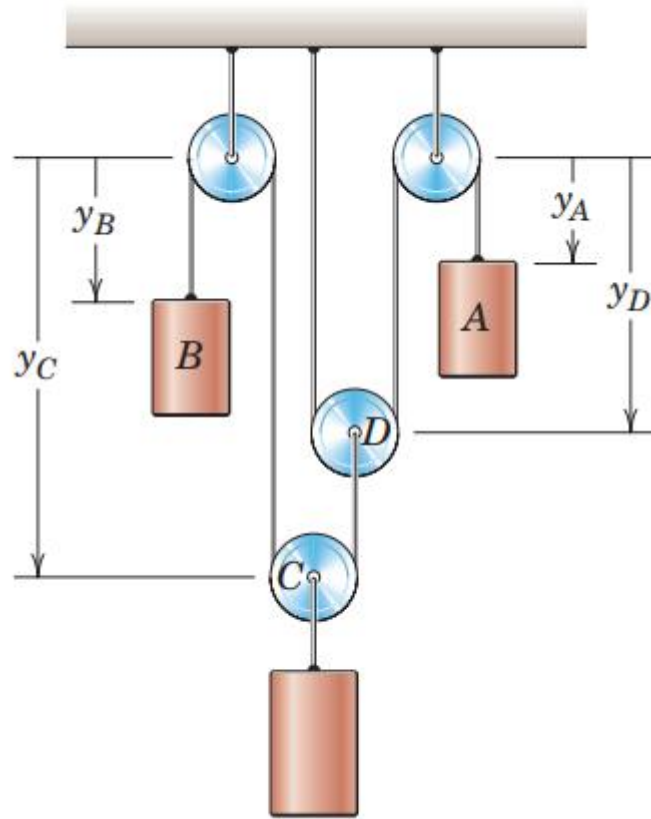
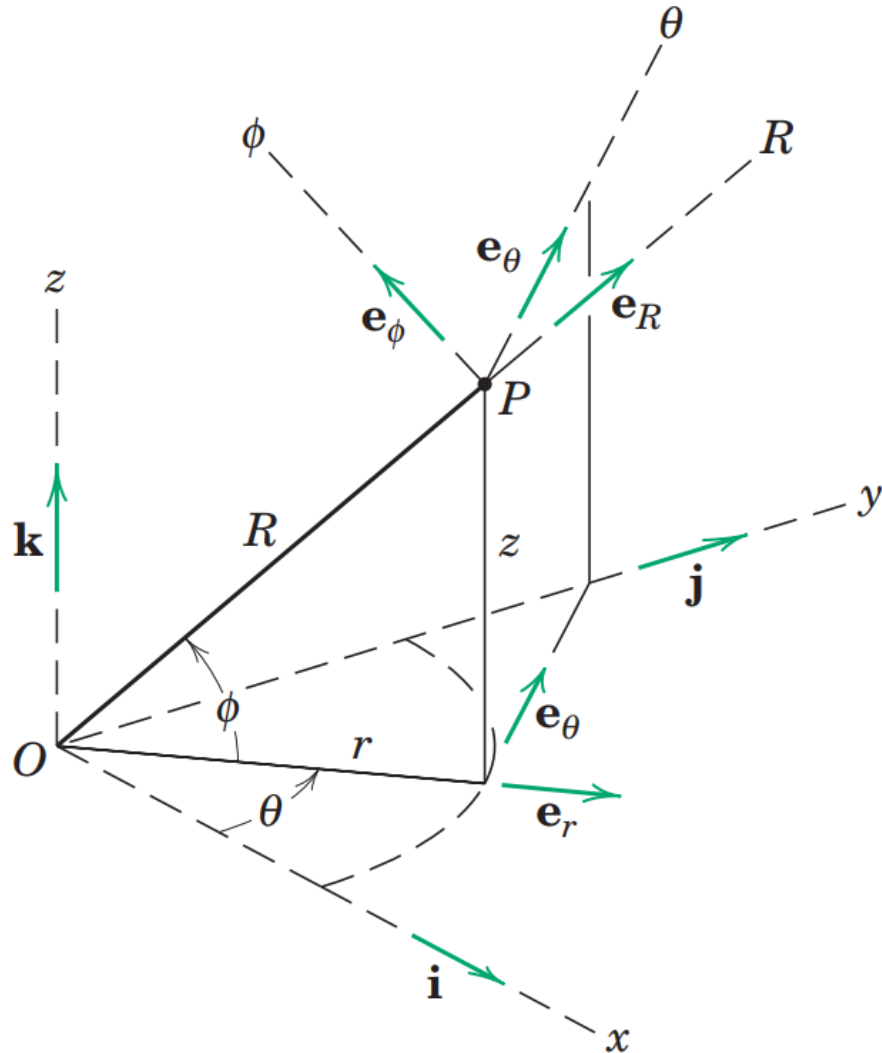


Figure: 7

Constrained motion of connected particles



Space Curvilinear Motion



Rectangular coordinate system

$$\mathbf{R} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

$$\mathbf{v} = \dot{\mathbf{R}} = \dot{x}\mathbf{i} + \dot{y}\mathbf{j} + \dot{z}\mathbf{k}$$

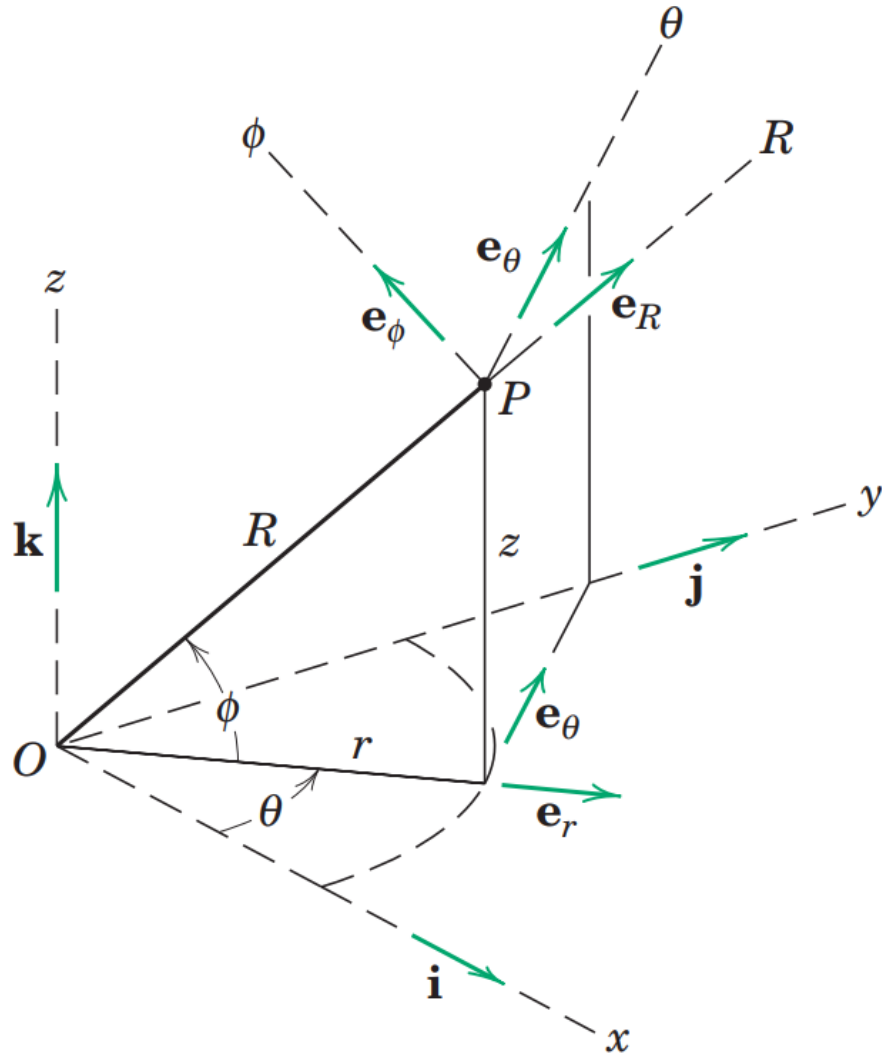
$$\mathbf{a} = \dot{\mathbf{v}} = \ddot{\mathbf{R}} = \ddot{x}\mathbf{i} + \ddot{y}\mathbf{j} + \ddot{z}\mathbf{k}$$

Cylindrical coordinate system

$$\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta + \dot{z}\mathbf{k}$$

$$\mathbf{a} = (\ddot{r} - r\dot{\theta}^2)\mathbf{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{e}_\theta + \ddot{z}\mathbf{k}$$

Space Curvilinear Motion



Spherical coordinate system

$$\mathbf{v} = v_R \mathbf{e}_R + v_\theta \mathbf{e}_\theta + v_\phi \mathbf{e}_\phi$$

$$v_R = \dot{R}$$

$$v_\theta = R \dot{\theta} \cos \phi$$

$$v_\phi = R \dot{\phi}$$

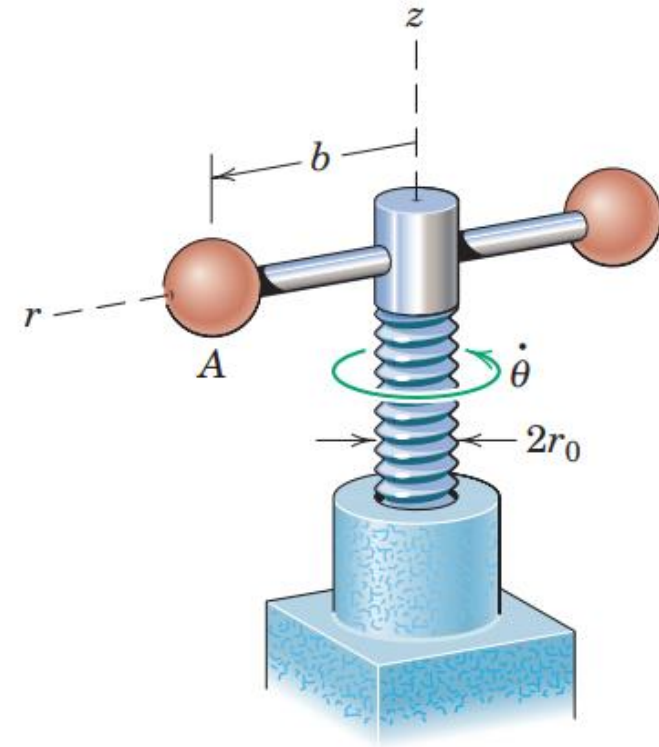
$$\mathbf{a} = a_R \mathbf{e}_R + a_\theta \mathbf{e}_\theta + a_\phi \mathbf{e}_\phi$$

$$a_R = \ddot{R} - R \dot{\phi}^2 - R \dot{\theta}^2 \cos^2 \phi$$

$$a_\theta = \frac{\cos \phi}{R} \frac{d}{dt} (R^2 \dot{\theta}) - 2R \dot{\theta} \dot{\phi} \sin \phi$$

$$a_\phi = \frac{1}{R} \frac{d}{dt} (R^2 \dot{\phi}) + R \dot{\theta}^2 \sin \phi \cos \phi$$

The power screw starts from rest and is given a rotational speed which increases uniformly with time t according to $\dot{\theta} = kt$, where k is a constant. Determine the expressions for the velocity v and acceleration a of the center of ball A when the screw has turned through one complete revolution from rest. The lead of the screw (advancement per revolution) is L .



9. The base structure of the fire-truck ladder rotates about a vertical axis through O with a constant angular velocity $\Omega = 10$ deg/s. At the same time, the ladder unit OB elevates at a constant rate $\dot{\phi} = 7$ deg/s, and section AB of the ladder extends from within section OA at the constant rate of 0.5 m/s. At the instant under consideration, $\phi = 30^\circ$, $\overline{OA} = 9$ m, and $\overline{AB} = 6$ m. Determine the magnitude of the velocity and acceleration of the end B of the ladder.

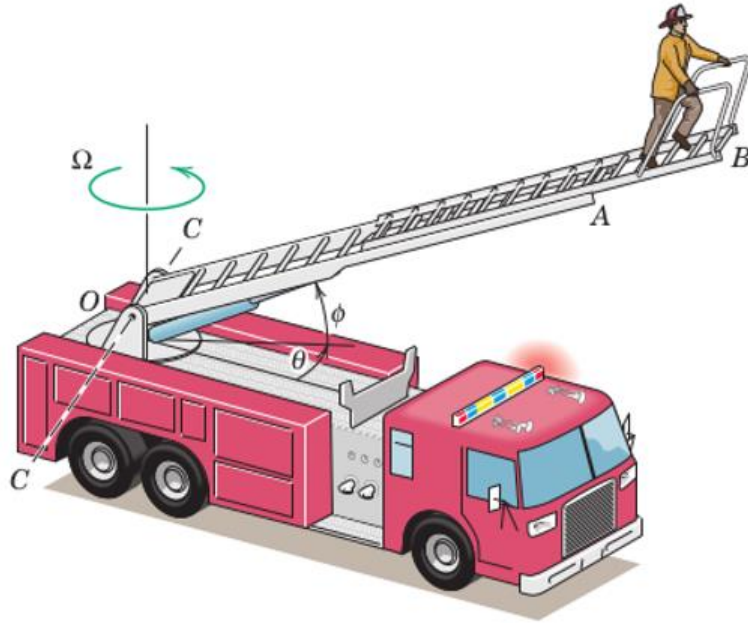


Figure: 9

Constrained motion of connected particles

8. If block B has a leftward velocity of 1.2 m/s, determine the velocity of cylinder A.

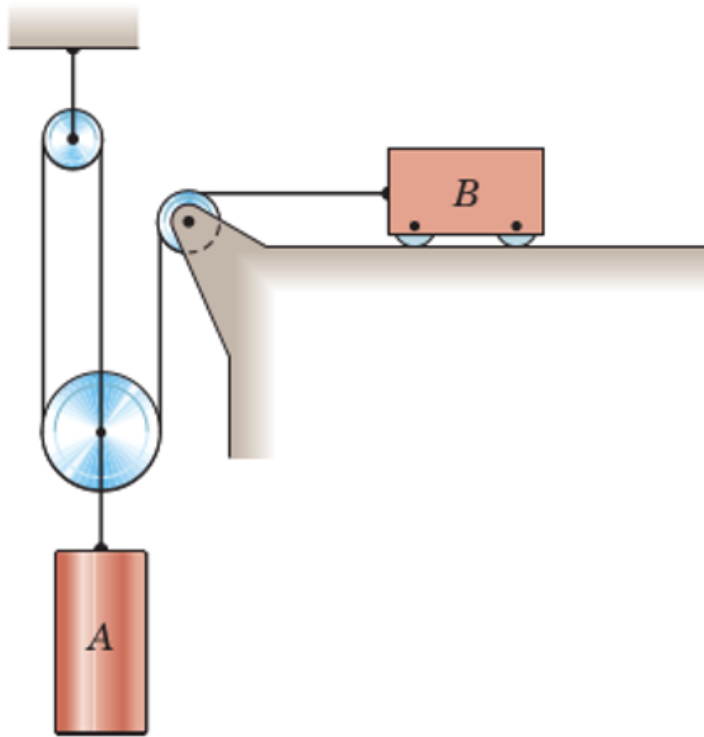


Figure: 8